GBA 6210

Case Study: Political Persuasion

Group 9:

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Background

- Predictive analytics in political campaigns involves using data to predict voter behavior and optimize campaign strategies.
- This case study is based on data provided by Ken Strasma, director of targeting for the Kerry and Obama campaigns.
- Our primary focus is to evaluate how campaign flyers influenced voter opinions using uplift modeling.

Objectives

- Assess the effectiveness of the campaign flyer in shifting voter opinions.
- Identify key predictor variables influencing voter opinion shifts.
- Apply predictive models and uplift modeling techniques to optimize campaign strategies.

Overview of Data Voter-Persuasion.csv

- **Target Variable:** MOVED_AD (binary: 1 if voter shifted towards the Democratic candidate, 0 otherwise).
- **Key Predictor Variables:**
  - MESSAGE_A: Binary variable indicating whether the voter received Message A (flyer promoting the Democratic candidate).
  - MESSAGE_A_REV: Binary variable indicating whether the voter received Message B (opposite of Message A).
  - Flyer: Combination of MESSAGE_A and MESSAGE_A_REV.
  - Other variables include AGE, party affiliation (PARTY_D, demographic information), gender (GENDER_F, and median household income (MED_HH_INC).

VOTER_ID	SET_NO	OPP_SEX	AGE	HH_...	HH_NR	HH_NI	MED_AGE	NH_WHITE	NH_AA
<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>

Effectiveness of the Flyer

Proportion with Message A:

40.24%

Proportion with Message B:

34.44%

```{r}
library(dplyr)
# Assuming 'MESSAGE_A' indicates receiving Message A (1 for yes, 0 for no)
# and 'MESSAGE_A_REV' indicates receiving Message B (1 for yes, 0 for no)

# Splitting the data into two groups based on the message type
data_message_a <- voter_data[voter_data$MESSAGE_A == 1,]
data_message_b <- voter_data[voter_data$MESSAGE_A_REV == 1,]

# Calculating the proportion of 'MOVED_A = 1' for each group
prop_message_a <- mean(data_message_a$MOVED_A)
prop_message_b <- mean(data_message_b$MOVED_A)

prop_message_a
prop_message_b
```

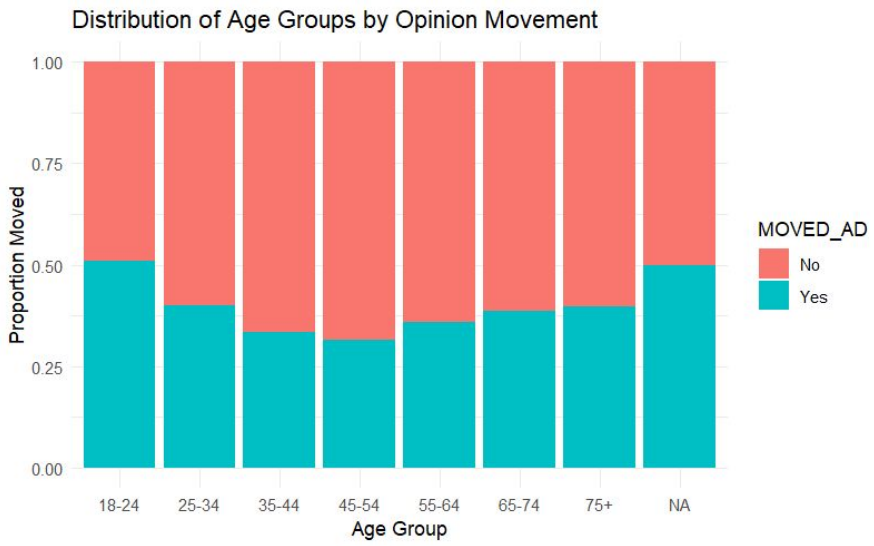
```
[1] 0.4024
[1] 0.3444
```

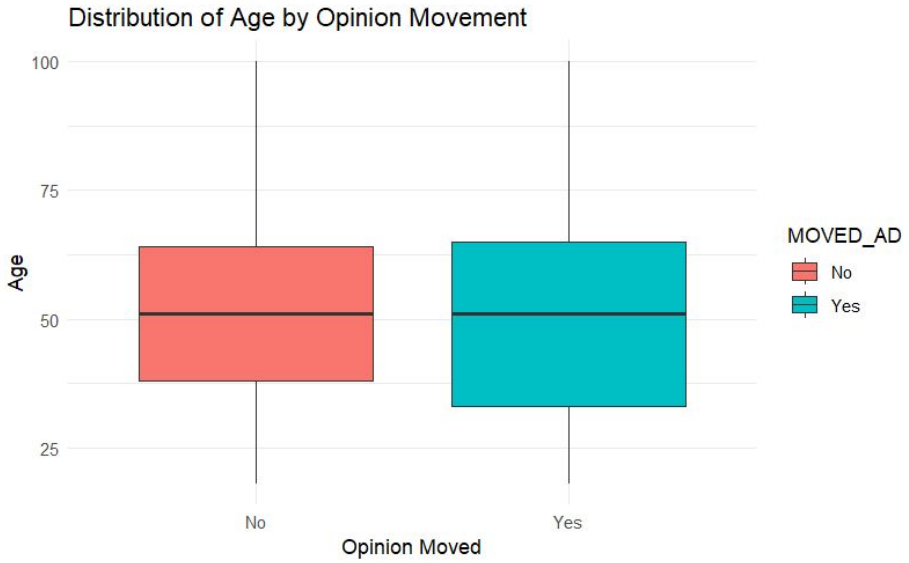
Visualization of Data

Out of a total of 37 columns, the 4 predictors we choose:

- Age
- Gender
- Democratic Party affiliation
- Income

Predictor - Age

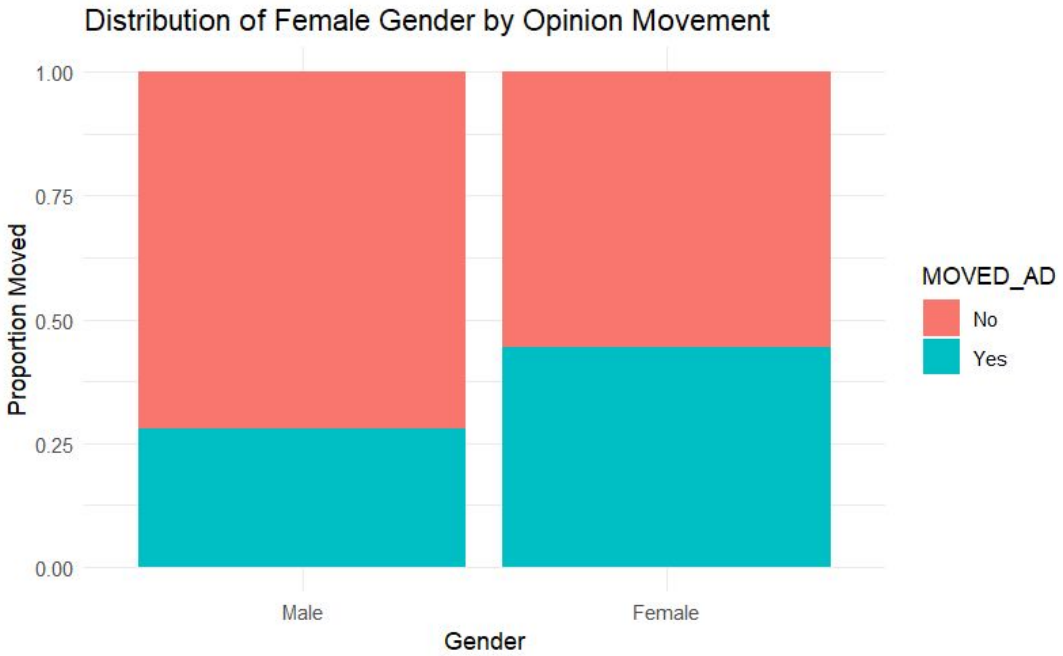




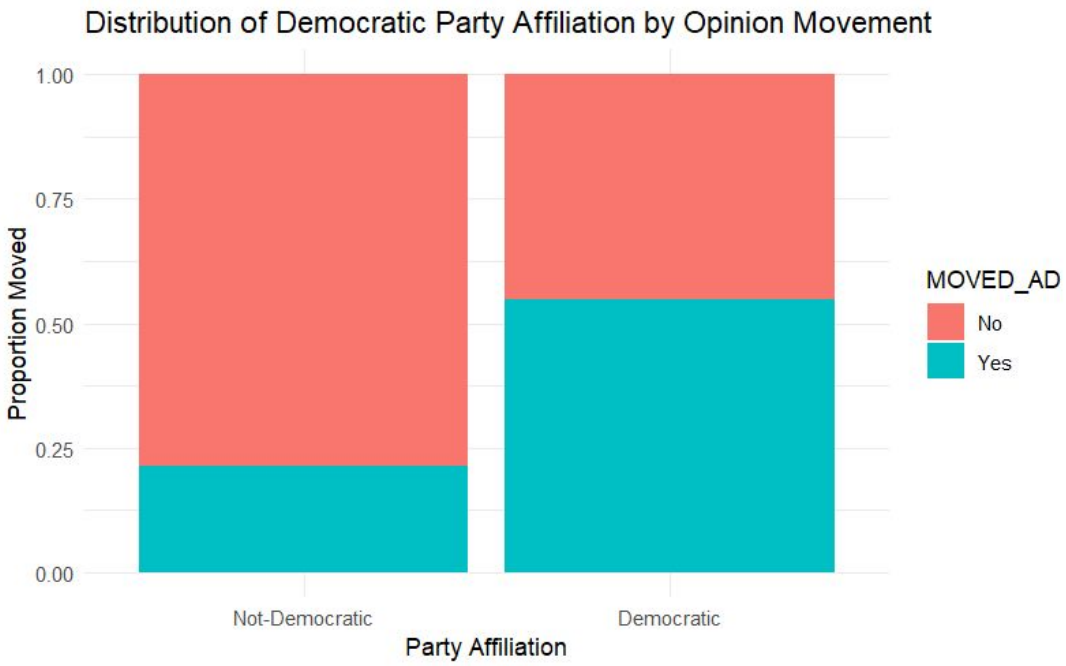
Younger voters more influenced by advertisements compared to older individuals. They tend to move towards the Democratic candidate more often.

Predictor - Gender

Female voters show a higher proportion of opinion shifts towards the Democratic candidate.

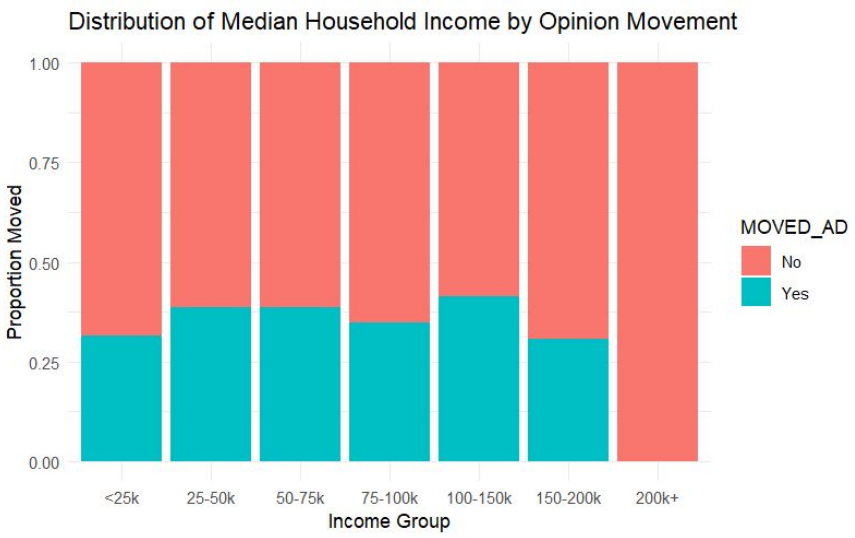


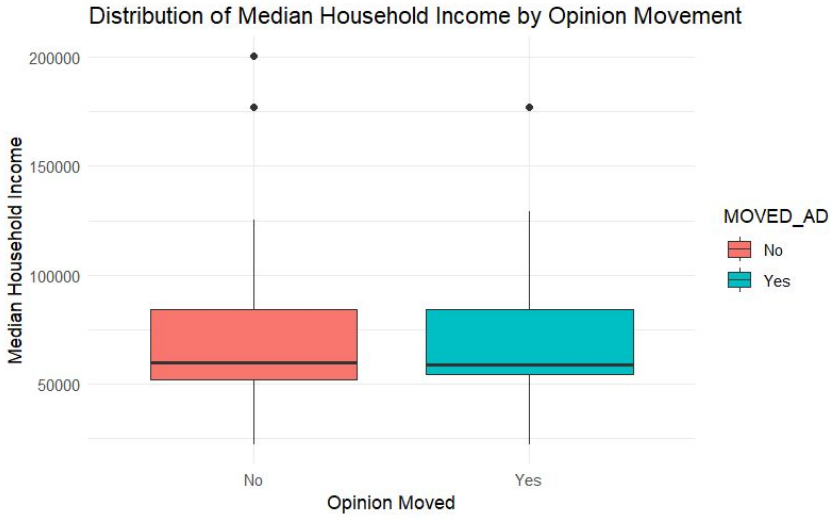
Predictor - Party Affiliation (Democratic)



Higher proportion of Democratic voters moved towards the Democratic candidate.

Predictor - Income





Voters with higher household income tend to move less towards the Democratic candidate.

Methodology - Models

1. **Logistic Regression:** provides a clear baseline for binary classification problems with strong interpretation.
2. **Classification Tree:** Offers interpretable decision rules, breaking down data splits visually and logically.
3. **Random Forest:** Enhances accuracy by combining multiple decision trees through ensemble learning to stabilize predictions.

Party affiliation (PARTY_D) is the most influential variable.

Methodology - Random Forest

Why Chosen:

Highest AUC

With an AUC of 0.7729, the random forest model outperforms both the logistic regression and decision tree models.

Robustness

Combines predictions from multiple trees, leading to improved accuracy and stability.

Feature Importance

Provides insights into the most important predictors for voter persuasion.

```
Logistic Regression AUC: 0.7226836
Classification Tree AUC: 0.7079589
Random Forest AUC: 0.7729733
```

* AUC (Area Under the Curve) measures the model sensitivity and specificity

Analysis and Result

Predictions Using the Best Model -Random Forest

Record 1 showed a 55.23% probability of moving towards the Democratic candidate, compared to 22.08% for Records 2 and 3.

```
predictions <- predict(tree_fit, newdata =
validation_data[1:3,], type = "prob")
```

	No <dbl>	Yes <dbl>
1	0.4476950	0.5523050
2	0.7792291	0.2207709
3	0.7792291	0.2207709

Derived Variable Analysis (Flyer-Reversed)

Flyer-Reversed increased the propensity for Record 1 to 75%, suggesting that modifying campaign messaging could significantly influence voter opinions.

```
    No  Yes
1 0.22 0.78
2 0.28 0.72
3 0.94 0.06
attr(,"class")
[1] "matrix" "array"  "votes"
```

Analysis and Result

Uplift Computation

- Record 1: Uplift = -0.062 (Negative)
- Record 2: Uplift = -0.026 (Negative)
- Record 3: Uplift = 0.024 (Positive)
- Records 1 and 2 exhibit negative uplift (flyer counterproductive).
- The flyer had a counterproductive effect on some voters.

Resource Allocation Strategy

Determine an uplift cutoff for targeting the top 10% of voters who would benefit from receiving the flyer.

- Cutoff Score: 0.0 (90th Percentile)
- Strategy: Target only voters with positive uplift scores (top 10%).

Conclusion

In this case study, we carried out voter persuasion and campaign strategy optimization through predictive analysis and promotion model.

By applying advanced models and gaining actionable insights, political campaigns can make data-driven decisions that improve voter targeting, increase campaign efficiency, and maximize resources in future electoral efforts.

Thank you!

Q&A?